

A Negative Answer to Exchangeable Joint-Mix Optimality for Convex Costs under Subset Uncertainty

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Abstract

Koike, Lin and Wang asked whether, under homogeneous marginals, an exchangeable joint mix solves their robust multi-marginal transport problem for a general convex cost. This note gives a finite counterexample. The marginal law is the uniform distribution on $\{0, 1, 2\}$, which is 3-completely mixable, and the uncertainty set is the symmetric fixed-cardinality class in which the unknown subset has size two. For

$$f_\varepsilon(x) = (x - 2)_+ + \varepsilon x^2, \quad 0 \leq \varepsilon < \frac{1}{2},$$

the robust optimum is computed exactly and is strictly smaller than the value of the exchangeable joint mix. Thus the answer to the question, as stated for general convex costs, is negative. This is a solution in the form of a counterexample, not merely partial progress. For $\varepsilon > 0$, the counterexample uses a strictly convex cost.

1 The robust transport problem

Let $[n] = \{1, \dots, n\}$. For a random vector $X = (X_1, \dots, X_n)$ and a subset $K \subseteq [n]$, write

$$S_K(X) = \sum_{i \in K} X_i.$$

Given marginal distributions $F_1, \dots, F_n$, a convex cost f , and an uncertainty set $\mathcal{M} \subseteq \mathcal{P}(2^{[n]})$, the robust objective considered by Koike, Lin and Wang [1] is

$$R_f^{\mathcal{M}}(X) = \sup_{\mu \in \mathcal{M}} \sum_{K \subseteq [n]} \mathbb{E}[f(S_K(X))] \mu(K), \quad (1)$$

where the optimization is over all couplings satisfying $X_i \sim F_i$. Thus the corresponding problem is

$$\inf \{ R_f^{\mathcal{M}}(X) : X_i \sim F_i, \ i = 1, \dots, n \}. \quad (2)$$

This is problem (8) of [1].

A random vector X is a joint mix with marginals $F_1, \dots, F_n$ if $X_i \sim F_i$ and $X_1 + \dots + X_n$ is almost surely constant. If $F_1 = \dots = F_n = F$ and such a vector exists, then F is called n -completely mixable. A coupling is exchangeable if its law is invariant under all coordinate permutations.

Koike, Lin and Wang prove exchangeable/NCD joint-mix optimality for the quadratic cost under homogeneous marginals and symmetric uncertainty sets. They ask whether an exchangeable joint mix also solves (2) for a general convex cost. The example below shows that the answer is no.

2 The example

Set $n = 3$ and let

$$F = \frac{1}{3}\delta_0 + \frac{1}{3}\delta_1 + \frac{1}{3}\delta_2.$$

This distribution is 3-completely mixable: the uniform law on the six permutations of $(0, 1, 2)$ has all three marginals equal to F and has constant coordinate sum 3.

Let $\mathcal{M}_2$ be the fixed-cardinality uncertainty set

$$\mathcal{M}_2 = \left\{ \mu \in \mathcal{P}(2^{[3]}) : \mu(\{K \subseteq [3] : |K| = 2\}) = 1 \right\}.$$

Equivalently, $\mathcal{M}_2$ is the set of all probability measures supported on the three two-element subsets of $[3]$. It is symmetric under all permutations of $[3]$. For this uncertainty set,

$$R_f^{\mathcal{M}_2}(X) = \max_{1 \leq i < j \leq 3} \mathbb{E}[f(X_i + X_j)]. \quad (3)$$

We use the one-parameter family of convex costs

$$f_\varepsilon(x) = (x - 2)_+ + \varepsilon x^2, \quad 0 \leq \varepsilon \leq \frac{1}{2}. \quad (4)$$

For $\varepsilon > 0$, the cost is strictly convex.

Theorem 1. *For F , $\mathcal{M}_2$, and f_ε as above, with $0 \leq \varepsilon \leq 1/2$,*

$$\inf \left\{ R_{f_\varepsilon}^{\mathcal{M}_2}(X) : X_1, X_2, X_3 \sim F \right\} = \frac{2}{9} + \frac{44}{9}\varepsilon. \quad (5)$$

For $0 \leq \varepsilon < 1/2$, every exchangeable joint mix has robust value

$$\frac{1}{3} + \frac{14}{3}\varepsilon, \quad (6)$$

which is strictly larger than (5). Hence, for every $0 \leq \varepsilon < 1/2$, no exchangeable joint mix solves the robust transport problem (2). In particular, choosing any $0 < \varepsilon < 1/2$ gives a strictly convex counterexample.

Proof. Write

$$g_\varepsilon(a, b) = f_\varepsilon(a + b) = (a + b - 2)_+ + \varepsilon(a + b)^2.$$

For $x = (x_1, x_2, x_3) \in \{0, 1, 2\}^3$, define

$$B_\varepsilon(x) = \sum_{1 \leq i < j \leq 3} g_\varepsilon(x_i, x_j), \quad N_r(x) = \#\{i : x_i = r\}, \quad r = 0, 1, 2.$$

We first prove a pointwise lower bound. Let

$$L_\varepsilon(x) = -\frac{2 + 8\varepsilon}{3}N_0(x) + 4\varepsilon N_1(x) + \frac{4 + 40\varepsilon}{3}N_2(x). \quad (7)$$

We claim that

$$B_\varepsilon(x) \geq L_\varepsilon(x) \quad \text{for all } x \in \{0, 1, 2\}^3, \quad (8)$$

whenever $0 \leq \varepsilon \leq 1/2$. Since $B_\varepsilon(x)$ and $L_\varepsilon(x)$ depend only on the count vector (N_0, N_1, N_2) , it suffices to check the ten possible count vectors. The following table gives the difference $B_\varepsilon - L_\varepsilon$.

(N_0, N_1, N_2)	$B_\varepsilon - L_\varepsilon$
$(0, 0, 3), (3, 0, 0)$	$2 + 8\varepsilon$
$(0, 1, 2), (2, 1, 0)$	$\frac{4}{3} + \frac{10}{3}\varepsilon$
$(0, 2, 1), (1, 2, 0)$	$\frac{2}{3} + \frac{2}{3}\varepsilon$
$(0, 3, 0), (1, 0, 2), (2, 0, 1)$	0
$(1, 1, 1)$	$\frac{1}{3} - \frac{2}{3}\varepsilon$

All entries are nonnegative for $0 \leq \varepsilon \leq 1/2$, proving (8).

Let $X = (X_1, X_2, X_3)$ be any feasible coupling with $X_i \sim F$. The marginal constraints imply

$$\mathbb{E}N_0(X) = \mathbb{E}N_1(X) = \mathbb{E}N_2(X) = 1.$$

Taking expectations in (8) gives

$$\begin{aligned} \mathbb{E}B_\varepsilon(X) &\geq -\frac{2+8\varepsilon}{3}\mathbb{E}N_0(X) + 4\varepsilon\mathbb{E}N_1(X) + \frac{4+40\varepsilon}{3}\mathbb{E}N_2(X) \\ &= \frac{2+44\varepsilon}{3}. \end{aligned}$$

Therefore, by (3),

$$\begin{aligned} R_{f_\varepsilon}^{\mathcal{M}_2}(X) &= \max_{1 \leq i < j \leq 3} \mathbb{E}g_\varepsilon(X_i, X_j) \\ &\geq \frac{1}{3} \sum_{1 \leq i < j \leq 3} \mathbb{E}g_\varepsilon(X_i, X_j) = \frac{1}{3}\mathbb{E}B_\varepsilon(X) \geq \frac{2}{9} + \frac{44}{9}\varepsilon. \end{aligned} \tag{9}$$

This proves the lower bound in (5) for all feasible couplings.

Now define a coupling X^* by

$$\mathbb{P}(X^* = (1, 1, 1)) = \frac{1}{3}, \tag{10}$$

with probability $1/9$ assigned to each distinct permutation of $(2, 2, 0)$, and probability $1/9$ assigned to each distinct permutation of $(2, 0, 0)$. These probabilities sum to one. The law is exchangeable. For each coordinate, the probabilities of the values 0, 1, and 2 are all $1/3$, so $X_i^* \sim F$.

For a fixed pair, say $(1, 2)$, a direct calculation gives

$$\mathbb{E}[(X_1^* + X_2^* - 2)_+] = \frac{2}{9}, \quad \mathbb{E}[(X_1^* + X_2^*)^2] = \frac{44}{9}.$$

Indeed, for the hinge term only the permutation in which the fixed pair equals $(2, 2)$ contributes, and it contributes 2 with probability $1/9$. The square expectation is

$$\frac{1}{3} \cdot 4 + \frac{1}{9} \cdot 16 + \frac{2}{9} \cdot 4 + \frac{2}{9} \cdot 4 = \frac{44}{9}.$$

By exchangeability the same value holds for all three pairs. Hence

$$R_{f_\varepsilon}^{\mathcal{M}_2}(X^*) = \frac{2}{9} + \frac{44}{9}\varepsilon.$$

Together with (9), this proves the exact optimum (5).

It remains to compute the value of exchangeable joint mixes. Let $Y = (Y_1, Y_2, Y_3)$ be any joint mix with marginals F . Since $\mathbb{E}(Y_1 + Y_2 + Y_3) = 3$, the constant sum must be 3. Because the coordinates take values in $\{0, 1, 2\}$, the only count vectors with total coordinate sum 3 are

$$(N_0, N_1, N_2) = (0, 3, 0) \quad \text{and} \quad (N_0, N_1, N_2) = (1, 1, 1).$$

The marginal constraint gives $\mathbb{E}N_0(Y) = 1$. Since the first count vector has $N_0 = 0$ and the second has $N_0 = 1$, the first count vector has probability zero. Thus every joint mix is supported on permutations of $(0, 1, 2)$.

An exchangeable joint mix is therefore the uniform distribution on the six permutations of $(0, 1, 2)$. For any fixed pair under this distribution, the pair sum is uniformly distributed on $\{1, 2, 3\}$. Consequently

$$R_{f_\varepsilon}^{\mathcal{M}_2}(Y) = \mathbb{E}f_\varepsilon(Y_1 + Y_2) = \frac{1}{3} + \frac{\varepsilon}{3}(1^2 + 2^2 + 3^2) = \frac{1}{3} + \frac{14}{3}\varepsilon.$$

Finally,

$$\left(\frac{1}{3} + \frac{14}{3}\varepsilon\right) - \left(\frac{2}{9} + \frac{44}{9}\varepsilon\right) = \frac{1 - 2\varepsilon}{9},$$

which is positive exactly when $\varepsilon < 1/2$. Therefore no exchangeable joint mix is optimal for $0 \leq \varepsilon < 1/2$. $\square$

Corollary 1. *The proposed extension of exchangeable joint-mix optimality from quadratic costs to arbitrary convex costs is false. The failure occurs already for $n = 3$, finite-support homogeneous marginals, a symmetric fixed-cardinality uncertainty set, and strictly convex costs.*

Remark 1 (What the example does and does not show). *The optimizer X^* in (10) is exchangeable, but it is not a joint mix: its coordinate sum takes the values 2, 3, and 4 with positive probability. Thus the counterexample does not say that exchangeability itself is harmful. Rather, it shows that the joint-mix constraint, which is optimal for the no-uncertainty convex aggregation problem and for the quadratic uncertain-subset problem, can be strictly suboptimal for convex costs under subset uncertainty.*

Remark 2 (Status). *Theorem 1 gives a full negative solution to the question as stated in [1]: an exchangeable joint mix need not solve (2) for a general convex cost. The result is not merely partial progress. The exact optimum is identified for the hinge cost $\varepsilon = 0$ and, more strongly, for the strictly convex perturbations $0 < \varepsilon < 1/2$.*

References

- [1] Takaaki Koike, Liyuan Lin, and Ruodu Wang. Joint mixability and notions of negative dependence. *Mathematics of Operations Research*, 49(4):2786–2802, 2024. doi: 10.1287/moor.2022.0121. arXiv: 2204.11438.